Se-Min Ryan Lim

Assistant Professor, Kookmin University
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Research Interests

"Application-driven, cross-layer system architectural innovations considering emerging accelerators and their real-world implementations"

Hardware-Software Co-Design

Hardware Accelerators

Memory- and Storage-Centric Processing

Reconfigurable Computing

Professional Experience

Kookmin University
Seoul, South Korea

Assistant Professor
Sep. 2025 – Present

- Pioneering application-driven system architectures that sustain lasting performance improvement in the post-Moore's Law era through cross-layer optimization and heterogeneous computing innovations.
- Lead the System Architecture for Target Applications (SATA) Lab as principal investigator, supervise student research projects, and serve as an instructor for computer and system architecture courses.

Semiconductor Research Corporation (SRC)
Irvine, CA, USA

Research Scholar
Jan. 2023 – Jun. 2025

- Developed prototype CPU-FPGA heterogeneous systems for data clustering and bioinformatics.

University of California, Irvine
Irvine, CA, USA

Graduate Student Researcher
Sep. 2020 – Jun. 2025

- Developed prototype CPU-FPGA heterogeneous systems for neural networks, cloud platforms, compression, and SAT solving.

Korea University
Seoul, South Korea

Visiting Researcher
Nov. 2019 – Aug. 2020

- Implemented specialized neural networks for human activity recognition.

Korea University Research & Business Foundation
Seoul, South Korea

Researcher
Apr. 2019 – Oct. 2019

- Implemented end-to-end FPGA accelerators for neural network inference.

Korea University
Seoul, South Korea

Graduate Research Assistant
Mar. 2017 – Feb. 2019

- Implemented faster and more efficient digital ALUs, FPGA neural network accelerators, and specialized neural networks for human activity recognition.

Korea University
Seoul, South Korea

Undergraduate Research Assistant
Mar. 2015 – Feb. 2017

- Implemented special-purpose embedded systems and Internet of Things devices.

Education

University of California, Irvine
Irvine, CA, USA

Doctor of Philosophy in Computer Science — *Sep. 2020 – Jun. 2025*

- Advisor: Sang-Woo Jun
- Committee: Sang-Woo Jun, Nikil Dutt, and Hyoukjun Kwon
- Dissertation: Application-Driven Cross-Layer Optimizations for CPU-FPGA Heterogeneous Systems

Korea University
Seoul, South Korea

Master of Engineering in Electronics Engineering — *Mar. 2017 – Feb. 2019*

Korea University
Seoul, South Korea

Bachelor of Engineering in Electronics Engineering — *Mar. 2011 – Feb. 2017*

Publications

* Co-first authors; † Co-corresponding authors.

Conference and Journal Papers

[1] Doeon Kim*, Seo-Young Lee*, Youngshin Jeong, Young H. Oh, and **Se-Min Lim**, "Sway: A Unified Acceleration Architecture for Quantized Compact Selective State-Space Models on Low-Cost Edge FPGAs." *IEEE/ACM International Symposium on Low Power Electronics and Design **(ISLPED)***, 2027. **[Prepare to submit]**

[2] Zion Lee, Dhanishta Nathan, Sang-Woo Jun, and **Se-Min Lim**, "Uatu: A CPU-FPGA Heterogeneous System for Scalable SAT Solving." *IEEE Symposium on Field-Programmable Custom Computing Machines **(FCCM)***, 2027. **[Prepare to submit]**

[3] Sang-Jun Park, Sang-Woo Jun, and **Se-Min Lim**, "Metamorpho: A CPU-FPGA Heterogeneous System for Fast and Accurate Motif Discovery." Submitted to the *64th Design Automation Conference **(DAC)***, 2027. **[Under review]**

[4] Esmerald Aliaj, **Se-Min Lim**, and Sang-Woo Jun, "FMiner: CPU-FPGA Heterogeneous Framework for Frequent Itemset Mining and Beyond," *IEEE/ACM International Conference on Computer-Aided Design **(ICCAD)***, 2026.

[5] Seongyoung Kang, **Se-Min Lim**, and Sang-Woo Jun, "Lembas: Cost-Efficient Genome Alignment with External Memory and FPGA Acceleration," *International Symposium on Computer Architecture **(ISCA)***, 2026.

[6] **Se-Min Lim**, Seongyoung Kang, and Sang-Woo Jun, "Bancroft: Genomics Acceleration Beyond On-Device Memory," *International Conference on Parallel Architectures and Compilation Techniques **(PACT)***, 2025.

[7] Gongjin Sun, Seongyoung Kang, Jane He, **Se-Min Lim**, and Sang-Woo Jun, "Labidus: RISC-V Overlay with Streaming Asynchronous Custom Instructions," *IEEE International Conference on Application-specific Systems, Architectures, and Processors **(ASAP)***, 2025.

[8] **Se-Min Lim**, Esmerald Aliaj, and Sang-Woo Jun, "Durin: CPU-FPGA Heterogeneous Platform for Scalable Low-Dimensional Data Clustering," *IEEE International Conference on Big Data **(BigData)***, 2024.

[9] **Se-Min Lim**, Esmerald Aliaj, and Sang-Woo Jun, "Morbius: Platform-Adaptive Hardware Accelerator for Scalable Sequence Motif Discovery," *IEEE International Conference on e-Science **(e-Science)***, 2024.

[10] **Se-Min Lim** and Sang-Woo Jun, "FlexForge: Efficient Reconfigurable Cloud Acceleration via Peripheral Resource Disaggregation," *Design, Automation & Test in Europe Conference & Exhibition **(DATE)***, 2024.

[11] **Se-Min Lim** and Sang-Woo Jun, "MobileNets Can Be Lossily Compressed: Neural Network Compression for Embedded Accelerators," ***Electronics***, 2022.

[12] **Se-Min Lim**, Jooyoung Park, and Hyeong-Cheol Oh, "Low-Cost Method for Recognizing Table Tennis Activity," ***IEICE Transactions on Information and Systems***, 2019.

[13] **Se-Min Lim**, Hyeong-Cheol Oh, Jaein Kim, Juwon Lee, and Jooyoung Park, "LSTM-Guided Coaching Assistant for Table Tennis Practice," ***Sensors***, 2018.

Theses

[14] **Se-Min Lim**, "Application-Driven Cross-Layer Optimizations for CPU-FPGA Heterogeneous Systems," *PhD dissertation, University of California, Irvine*, 2025.

[15] **Se-Min Lim**, "AI-Based Coaching Assistant System for Sports Practice," *Master's thesis, Korea University*, 2019.

Research Funding

[1] **Co-Principal Investigator**, "Development of Advanced Technologies on SmartNICs," WINSTECHNET, US$80,000, Dec. 2025 – May 2026.

Open-Source Software

[1] **blueVitis**, github.com/SeMinLim/bluevitis. *Creator & Maintainer.* "An Advanced, High-Performance Boilerplate Codebase for AMD FPGA Kernel Development using Bluespec SystemVerilog (BSV)."

[2] **blueYosys**, github.com/SeMinLim/blueyosys. *Creator & Maintainer.* "The First Open-Source Boilerplate Codebase for Lattice Embedded FPGA Kernel Development using Bluespec SystemVerilog (BSV) and the Yosys Toolchain."

[3] **SIRVE**, github.com/SeMinLim/sirve. *Creator & Maintainer.* "An Interactive RV32I Emulator for Computer Architecture Education, Integrating a Built-In Assembler with Step-by-Step Execution and Register/Memory Inspection."

Research Advising & Mentoring

Undergraduate Interns

Sang-Jun Park — Kookmin University

Seo-Young Lee — Kookmin University

Zion Lee — Kookmin University

Dhanishta Nathan — R.V. College of Engineering

Teaching Experience

Kookmin University

Seoul, South Korea

Assistant Professor

Sep. 2025 – Present

- Computer Architecture (Fall 2025, Fall 2026)
- Practices in Software Convergence (Fall 2025, Fall 2026)
- C++ Programming (Fall 2025)
- Software Project I (Spring 2026)

University of California, Irvine

Irvine, CA, USA

Teaching Assistant

Sep. 2020 – Mar. 2025

- Principles of Operating Systems (Fall 2020)
- Computer Systems Architecture (Winter 2025)

Korea University

Seoul, South Korea

Teaching Assistant

Mar. 2017 – Dec. 2018

- Computer Architecture, Digital System Laboratory (Fall 2018)
- Electronic Circuit II, Electric Circuits II, Signals and Systems I (Spring 2018)
- Introduction to Applied Mathematics, Discrete Mathematics (Fall 2017)
- Pre-Calculus, Calculus (Spring 2017)

Honors & Awards

2026 **Best Paper Award** *Korea Computer Congress (KCC)*, Corresponding author

2018 **Best Poster Award** *Korea Computer Congress (KCC)*, First author

2016 **Best Research of Undergraduate Student Award** *Korea University*

Fellowships, Scholarships & Assistantships

University of California, Irvine
Graduate research and teaching assistantships — *Sep. 2020 – Jun. 2025*

Korea University
Graduate research and teaching assistantships — *Mar. 2017 – Feb. 2019*

Korea Student Aid Foundation
National Grant — *Jun. 2015 – Dec. 2016*

Korea University
Scholarship for Undergraduate Research Assistant — *Sep. 2016*

Korea University
Scholarship of Korea University Alumni Association — *Dec. 2015*

Professional Service

Organizing Committee *KIISE (2026)*
Program Committee *HPCAsia (2026)*
Session Chair *PACT (2025)*
External Review Committee *USENIX ATC (2024)*
Journal Reviewer *Journal of Big Data (2025); IEEE Systems Journal (2021, 2022)*

Talks

2026 **"Beyond Moore's Law: Breakthrough System Architectural Solutions for Next-Generation Heterogeneous Systems"** *Ajou University*
2026 **"Addressing the Data Movement Overhead in Genomic Accelerators"** *Computer System Society Conference*
2025 **"Addressing the Data Movement Overhead in Genomic Accelerators"** *Korea Software Congress*

Patents

[1] Hyeong-Cheol Oh, **Se-Min Lim**, and Seongjin Choi, "Pipelined Squarer for Unsigned Integers of up to 16 Bits," *Korean Patent No. 10-1974779*, filed Apr. 16, 2018; registered Apr. 25, 2019.

Technical Skills

Programming Language Skills

- Hardware languages: Bluespec SystemVerilog (BSV), Verilog, VHDL.
- Programming: C++, C, Python, MATLAB, HTML; bare-metal programming.
- Computing frameworks: OpenCL, CUDA, PyTorch, Keras, TensorFlow.

Software & Program Skills

- FPGA development: AMD/Xilinx Vivado and Vitis, Intel Quartus Prime, Xilinx Runtime (XRT), Bluespec Compiler and Bluesim, Yosys, nextpnr.
- Systems and development tools: Linux, Kubernetes, Docker, Multisim, Visual Studio, PyCharm, Jupyter, Spyder.

Languages

English

- Professional working proficiency

Korean

- Native proficiency